
A comparative analysis of Von-Heijne, SVM and PLM Models for the detecton of Signal Peptides.

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Abstract:

Motivation: Signal peptides enable proteins to enter the secretory pathway. Given their importance and the high cost of in-vivo assays, accurate in-silico SP detection models have taken on an important role in bioinformatics.

Results: With this paper, we provide three models for the detection of Signal Peptides from sequence information. The first, based on the Von-Heijne algorithm, produces a relatively accurate classifier, with an MCC of 0.66. The second model is based on Support Vector Machines and achieves an higher MCC of 0.86 and f_1 score of 0.87, supporting an overall stronger model. Given the complexity of the full SVM model, we also trained a reduced model, which trades some performance (MCC of 0.73, and f_1 of 0.76), for a significantly lower number of input features, lowering the amount of preprocessing needed and the overall complexity of the final model. We also present a predictor based on Protein Language Models, which behaves as a near perfect classifier, with an MCC of 0.96 and f_1 score of 0.96, but is not directly comparable to the other two models, given the much higher complexity of the underlying method.

Code Availability: The full source code for this paper is available at <https://github.com/FabrizioG202/lab2>.

1. Introduction

In both eukaryotic and prokaryotic cells, for a protein to enter the secretory pathway, it must contain a specific targeting signal. Often, this signal is a short sequence located at the N-terminus of proteins [1]. The signal peptide (SP) has a distinct three-domain structure (depicted in Figure 1) with a positively charged N-terminal region (N-region), a central hydrophobic α -helical core (H-region) and a more polar C-terminal region (C-region) containing the cleavage site [1]. After completing its mission as protein ‘zip code’, the short peptide is cleaved by a membrane-bound peptidase, which separates it from the mature protein [2]. Amino acid substitutions in the C-region revealed a universally conserved recognition site for the leader peptidase, with the three-residue motif $\text{Ala}^{-3}\text{-X-Ala}^{-1}$, where X can be any amino acid and numbering is relative to the cleavage site [3].

Given their importance in many areas of molecular cell biology such as targeted drug design [4], the accurate detection of SPs remains a crucial

task in bioinformatics and has been addressed using a variety of approaches, including machine learning (SVMs [5], [6] and Bayesian networks [7], Hidden Markov Models [8]), and, more recently, deep learning [9].

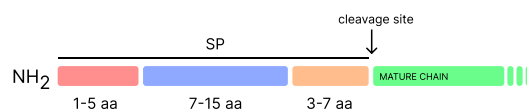


Figure 1: Signal peptide structure, adapted from [1]

In this work, we propose a comparative analysis of different approaches for the detection of SPs from protein sequences. We compare the performance of the Von-Heijne model, which is based on a position-specific weight matrix (PSWM) generated from the context around the cleavage site, with that of a Support Vector Machine (SVM) model trained on a variety of features extracted from the sequence and a simple Multi-Layer Perceptron (MLP) trained on sequence embeddings computed using a PLM.

2. Materials and Methods

2.1. Data Collection

Reviewed, non-fragment, protein sequences from eukaryotic taxa were queried from UniprotKB [10], and collected into 2 distinct sets: positive and negative. For the positive set, we selected sequences with experimental evidence for the presence of a signal peptide (ft_signal_exp:*) For the negative set, we selected proteins not annotated with a signal peptide but having experimental evidence for subcellular localization in non-secretory compartments, namely nucleus (cc_scl_term_exp:SL-0191), peroxisome (cc_scl_term_exp:SL-0204), cell membrane / plasma membrane (cc_scl_term_exp:SL-0039), cytosol (cc_scl_term_exp:SL-0091), plastid (cc_scl_term_exp:SL-0209) or mitochondrion (cc_scl_term_exp:SL-0173). For sequences in the negative set, we gathered information about the presence of transmembrane (TM) helices, whose hydrophobicity profile mimicks those of signal peptides.

Further selection on the positive entries was carried out in order to select proteins which had informaton about the cleavage site and a Signal Peptide with a length of at least 14 residues.

To perform an initial screening for possible differences between the two sets, we plotted the distribution of sequence lengths in the positive and negative datasets (Figure 2), which showed similar trends among the two groups.

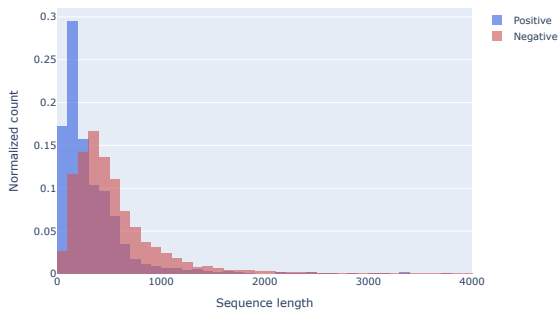


Figure 2: Length distribution of positive and negative sequences

To ensure equal distribution among the kindoms, the composition of each set was considered (Figure 3).

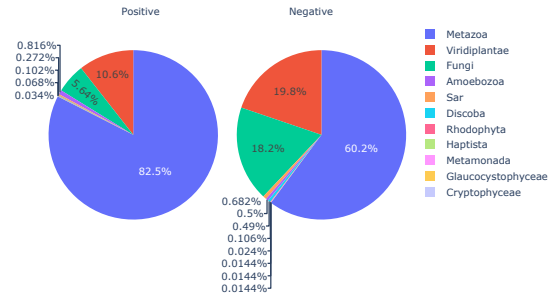


Figure 3: Kingdom distribution of positive and negative datasets

The distribution of cleavage-site positions, corresponding to the length of the cleaved N-terminal region in positive sequences, is shown in Figure 4. The distribution was used to quickly scan for outliers and to crudely confirm the biological plausibility of the information.

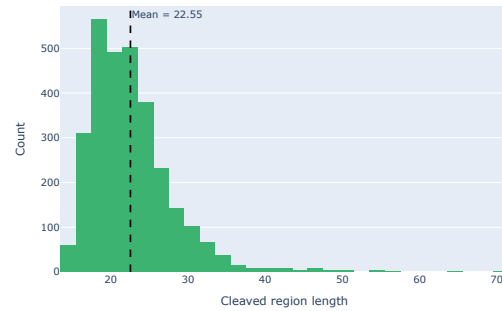


Figure 4: Length distribution of the cleaved region in positive sequences

2.2. Clustering

Sequences were clustered using mmseqs2 [11], using parameters `--min-seq-id 0.3, cluster-mode 1 -cov-mode 0 -c 0.4`, in order to cluster sequences with a similarity of at least 30% and an alignment spanning at least 40% of the total sequence length. Then, only representative sequences from each cluster were selected for downstream analysis. The number of sequences, before and after clustering is reported in Table 1.

Dataset	Before Clustering	After Clustering
Positive	2942	1093
Negative (total)	20806	9028
Negative (w/o TM)	18301	8113
Negative (with TM)	2505	915

Table 1: Dataset sizes before and after clustering

2.3. Von-Heijne Model

The full dataset was split into a training and a test set with a ratio of 0.8. A Position-Specific Weight Matrix (PSWM) was then generated using 16-residue wide windows (13aa upstream to 2aa downstream) surrounding the cleavage site. Amino acid frequencies were normalized using background amino acid frequencies from SwissProt. After initializing the matrix with pseudo-counts of 1, the weights are computed as follows:

$$w(a, i) = \log_2 \left(\frac{f(a, i)}{f(a)} \right)$$

where $f(a, i)$ is the frequency of amino acid a at position i in the training set, and $f(a)$ is the background frequency of amino acid a in SwissProt. The score for a given sequence is then computed as:

$$S = \max_{j=1}^{90-16+1} \sum_{i=1}^{16} w(a_{j+i-1}, i)$$

Scores are computed over the first 90 amino acids of the given sequence, using a sliding 16-residue window. A sequence is labeled as positive for an SP if the Score S is greater than a threshold t .

$$\text{label} = \begin{cases} \text{positive} & \text{if } S > t \\ \text{negative} & \text{if } S \leq t \end{cases}$$

The optimal threshold for the dataset was computed using 5-fold cross validation. For each fold, the data was split into validation and test sets. On the validation set, we computed precision and recall values across all possible threshold values, then selected the threshold that maximized the f_1 score (see Section 2.6). This optimal threshold was then evaluated on the corresponding test set to compute performance metrics. The final

threshold reported is the average of the optimal thresholds across all 5 folds.

2.4. Feature Extraction

To build a model that can handle sequence data, sequences must be encoded as numerical representation. There are many possible ways of extracting meaningful numerical features from a sequence, for both the full and reduced SVM model, we employed 2 types of features:

- Amino acid composition up to a cut-off k (n_term_composition) and from k to an hardcoded limit N (set to 90) (c_term_composition)
- Protein Scales computed over the first N aminoacids with a window size of 5 (Figure 5). The hydrophobicity scale ([12]), alpha-helix tendency scale ([13]), transmembrane tendency scale ([14]), bulkiness scale ([15]), and polarity scale([15]) were used. For each scale, the minimum, maximum, average, standard deviation and position of maximum were used as features.

For this experiment, k was chosen to be the average position of the cut site, roughly 22 aminoacids from the N-Terminus.

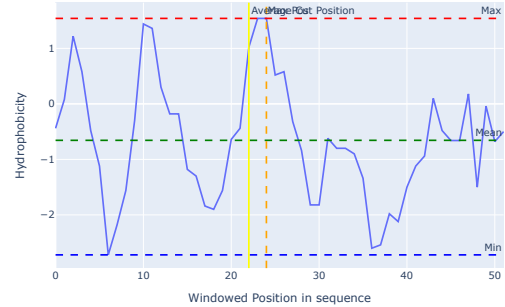


Figure 5: Example of feature extraction for a given sequence. Protein scales are computed on the first N aminoacids with a window size of 5

2.5. Support Vector Machine

Support Vector Machines (SVM) are a machine learning methodology that can be used for two-group classification problems. SVMs separate classes using an hyperplane, defined by a small set of the input data points called support vectors. Albeit trivial for linearly separable classes, vector machines can make use of a kernel function, which allows the mapping of input points from

the original space to an higher-dimension space (feature space). In this new space, a linear decision boundary can be established, which, when mapped back to the original space, will produce a complex, non-linear decision boundary, enabling correct classification of complex classes [16]. A commonly used kernel function is Radial Basis Function (RBF), which is defined as follows:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

The γ parameter defines how much influence a single training example has. A low γ value means that the influence of a single training example is far-reaching, while a high γ value means that the influence is limited to close neighbors. For both linear and RBF kernels, the C parameter controls the trade-off between maximizing the margin and minimizing the classification error, with low C values leading to a wider margin and a simpler model, and high C values leading to a narrower margin, with few training errors but a more complex model.

We used a grid search to find optimal hyperparameters, testing both RBF and linear kernels. For the RBF kernel, we evaluated C values of 0.1, 1, 2, 4, and 8, combined with gamma values of “scale”, 0.001, 0.01, 0.1, 1, and 2. For linear kernels, we tested C values of 0.1, 1, 2, 4, and 8. Each model was fitted and evaluated on an 80% training split. The model yielding the highest f_1 score was found to have a rbf kernel, with $C = 2$ and $\gamma = 0.01$, with a mean f_1 score across the 5 folds ran on the training set of 0.87, and was thus selected for downstream analysis.

2.6. Evaluation and comparison

Rather than relying on accuracy alone, since the dataset is greatly imbalanced, to evaluate and compare the performance of the models, we computed several metrics on the test set. The metrics were calculated based on the following formulas:

Accuracy: The proportion of correctly classified instances among all instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision: The proportion of true positive predictions among all positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity): The proportion of true positive instances that were correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN}$$

f_1 Score: The harmonic mean of precision and recall.

$$F_1 = 2 \cdot \frac{(\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}$$

Matthews Correlation Coefficient (MCC): A balanced measure that takes into account all four confusion matrix categories.

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Where TP (True Positives), TN (True Negatives), FP (False Positives), and FN (False Negatives) are the values from the confusion matrix.

3. Results

3.1. SP Motifs

From all SP-endowed sequences, a motif logo (Figure 6) was generated using the context around the cleavage site (13aa upstream to 2aa downstream). The motif shows the predominance of small and neutral residues, such as Alanine, in positions -1 and -3 , which has already been noted in previous literature [17] and is consistent with the suggested Ala⁻³-X-Ala⁻¹ signature. Moreover, the high frequency of hydrophobic residues (Leucine, Valine, Phenylalanine) is consistent with the structure of SPs presented above, particularly the H-Region, which holds an hydrophobic core.

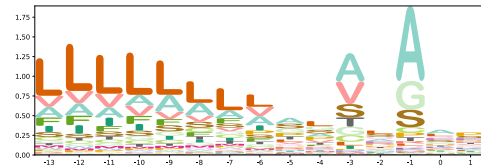


Figure 6: Motif logo for the context (-13aa to 2aa) surrounding the cleavage site, showing the conserved Ala⁻³-X-Ala⁻¹ motif. The cleavage site is between residues -1 and 0

3.2. Von Heijne Model

For the Von-Heijne model, the computed optimal threshold was 9.25, confusion matrix computed on the test set is reported in Figure 7. The

model achieved an accuracy of 0.9378, precision of 0.7474, recall of 0.6532, f_1 score of 0.6971, and MCC of 0.6645. The model reaches an high accuracy, but performance on the positive class is limited, detecting only around 65% of actual positives. This is likely due to both the simplicity of the model, which makes it unable to capture long-range or high-level patterns and the limited nature of the input data, which is based on a small window around the cleavage site, which may not capture all the relevant information for SP detection.

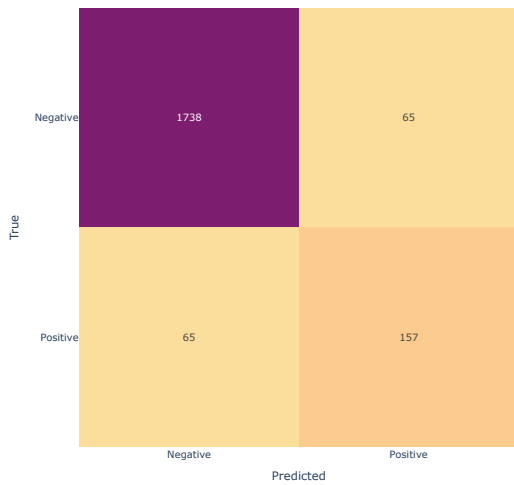


Figure 7: Confusion matrix for the Von-Heijne model

3.2.1. Analysis of False Positives

We analysed the set of false Positives (wrongly classified as having a Signal Peptide) of the von-heijne method and as expected, we found it to be significantly enriched (Fisher’s exact test, one-sided, $OR = 6.45$, $p\text{-value} = 9.45 \times 10^{-11}$) for proteins having a TM helix. This is likely due to the fact that the hydrophobicity profile of a TM helix mimicks that one of a Signal Peptide.

3.3. Support Vector Machine

The model, as chosen in the hyperparameter tuning phase, is a SVM with RBF kernel, $C = 2$ and $\gamma = 0.01$. The confusion matrix computed on the test set is reported in Figure 8. The model achieved an accuracy of 0.9743, precision of 0.9034, recall of 0.8539, f_1 score of

0.8779, and MCC of 0.8640. The SVM model thus outperforms the Von-Heijne model across all metrics, showing particularly strong improvement in recall (85.39% vs 65.32%), indicating better detection of signal peptides. This means that the SVM model is both more reliable when predicting positives and much better at detecting actual positive cases. It is worth noting that this increase in performance mirrors a great increase in complexity of the underlying method, which is structurally more intricate than a simple PSWM, and also requires far greater preprocessing to compute the features. To mitigate this, we tried producing a reduced model, using only the top 5 features selected using permutation importance and the top 5 features selected using the feature importance of a Random Forest Classifier, which we trained on the same dataset.

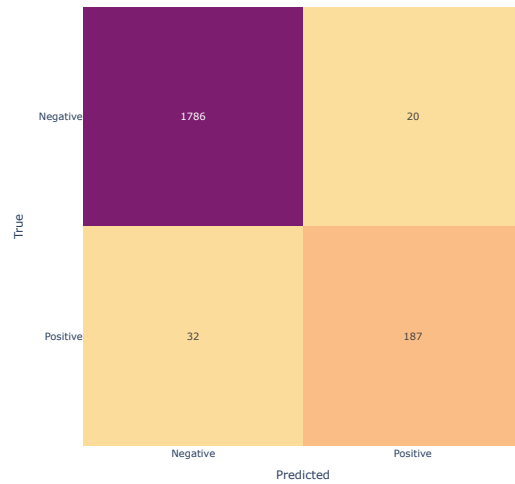


Figure 8: Confusion matrix for the SVM model

3.3.1. Feature Importance

In order to understand which features contribute the most to the ability of our model to classify sequences, we employed two methods. As a first test, we assessed permutation importance, where values of each feature are randomly shuffled in order to break their relationship with the target label, using the drop in the classifier accuracy as a proxy for the importance of the feature. As a second test, we trained a Random Forest Classifier with 500 classifiers and exploited the model’s own feature importance, which is computed as a result of the training process. We also evaluated

the performance of the resulting random forest classifier, which itself proved to be a good model, with an accuracy of 0.9580, precision of 0.9036, recall of 0.6849, f_1 score of 0.7792, and MCC of 0.7654.

The top 5 features identified by each method are shown in Table 2 and Table 3. Both methods consistently identified `n_terminal_comp_L` (leucine composition in the N-terminal region) and `tm_max` (maximum transmembrane tendency) as important features, albeit with different ranks. The two methods of detecting feature importance differ with respect to the measured metrics, but the presence of the attributes in both rankings strengthens confidence that these are genuinely informative. Leucine Composition’s importance is consistent with signal peptides containing a hydrophobic core, where Leucines, and more generally hydrophobic residues, such as Alanine are commonly enriched. Furthermore, the presence of `n_terminal_comp_D`, a hydrophilic amino acid, may be representative of the fact that the model is informed by its general depletion.

Feature	Importance Mean
<code>n_terminal_comp_L</code>	0.068479
<code>n_terminal_comp_A</code>	0.010622
<code>tm_max</code>	0.008453
<code>kd_max_pos</code>	0.008225
<code>n_terminal_comp_D</code>	0.008195

Table 2: Top 5 features by permutation importance

Feature	Importance
<code>kd_max_pos</code>	0.11984
<code>tm_max_pos</code>	0.105942
<code>tm_max</code>	0.091583
<code>kd_max</code>	0.078178
<code>n_terminal_comp_L</code>	0.06908

Table 3: Top 5 features by random forest importance

3.3.2. Reduced SVM Model

We then trained a reduced model using the combination of the 5 most important features found by each method. The 7 chosen features were:

- `kd_max`: Maximum hydrophobicity value

- `kd_max_pos`: Position of maximum hydrophobicity
- `n_terminal_comp_A`: Alanine composition in the N-terminal region
- `n_terminal_comp_D`: Aspartic acid composition in the N-terminal region
- `n_terminal_comp_L`: Leucine composition in the N-terminal region
- `tm_max`: Maximum transmembrane tendency
- `tm_max_pos`: Position of maximum transmembrane tendency

The architecture for the reduced model was selected using the same method used for the full model, including the same pool of possible classifiers. The reduced model achieved an accuracy of 0.9521, precision of 0.8177, recall of 0.7169, f_1 score of 0.7640, and MCC of 0.7394. The confusion matrix computed on the test set is reported in Figure 9. While this represents a decrease in performance compared to the full SVM model (f_1 score of 0.8779 vs 0.7640), the reduced model still outperforms the Von-Heijne model (f_1 score of 0.6971) while using only 7 features instead of the full feature set.

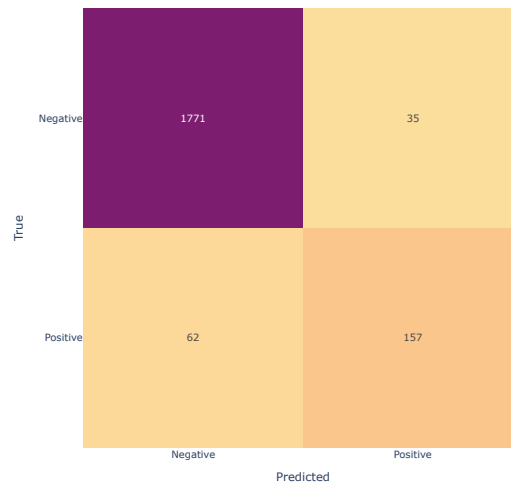


Figure 9: Confusion matrix for the reduced SVM model

4. MLP-based classifier

Protein Language models (PLMs) are adaptations of Large Language Models (LLMs) which treat amino acid sequences as sentences. They are trained on millions of sequences and are able

to learn extremely complex patterns, which can be informative for a wide range of tasks, such as structure prediction, annotation, and classification [18]. We trained a simple Multi-layer Perceptron with an hidden layer of 100 neurons on sequence embeddings computed on the first 90 residues of each sequence. The limited size of the input dataset precludes training a model from scratch. For this reason, the Ankh Protein Language Model [19] was chosen as both the tokenizer and embedder. The model size chosen was the base one, which despite its limited size, achieves performance similar or surpassing those of larger models, such as ESM and ESM-2 [20], [21]. The generated embeddings have dimensionality 768. The MLP model achieved exceptional performance with an accuracy of 0.9926, precision of 0.9766, recall of 0.9543, f_1 score of 0.9654, and MCC of 0.9613. The confusion matrix is reported in Figure 10. This model is the most powerful among the ones presented in this work. However, it is by far the most complex, both for the classifier and for the preprocessing step, where the embeddings are computed, requiring extensive computational capacity. This model is thus not directly comparable to the others, but it is worth noting that the use of PLM embeddings allows it to capture complex sequence patterns and relationships that may be difficult to capture with traditional feature engineering approaches, which likely contributes to its superior performance.

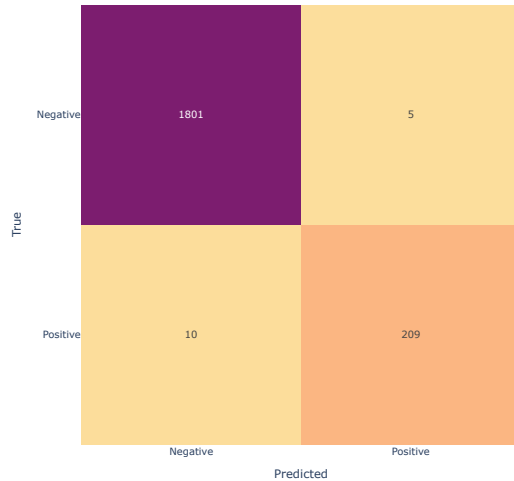


Figure 10: Confusion matrix for the MLP model

5. Conclusion

The results show the existence of a clear trade-off between simplicity and performance. The Von-Hejne predictor, despite its simplicity, reaches an acceptable accuracy, but is plagued by a limited recall, meaning it fails at recognizing a significant portion of actual positives. Both models based on Support Vector Machines display an increased overall performance, with the full model reaching an f_1 score of 0.8779, and the reduced model reaching an f_1 score of 0.7640. This performance increase is likely due to the richer nature of the input features, which nicely describe physicochemical features that sequence information alone fails to capture. The reduced model, while showing a decrease in performance compared to the full model, still outperforms the Von-Heijne model, while using only 7 features instead of the full feature set, which makes it a good candidate for applications where computational resources are limited. Another aspect of the comparison that stands out is the loss in interpretability that exists with larger and more complex models. The very structure underpinning the von heijne model, that of a positional matrix, makes it biologically interpretable, as it allows to easily identify conserved positions and amino acid, and can be related to known peptidase recognition motifs. On the other hand, the SVM models are more difficult to interpret, as they rely on a

complex combination of features and non-linear decision boundaries.

The extremely high accuracy of the MLP model suggests how the use of Protein Language models can capture high-complexity relationships spanning the whole input sequence, which cannot be encoded by hand-crafted features.

6. Code Availability

The code used for this analysis is available at: <https://github.com/FabrizioG202/lab2>

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